BENZIR AHAMMED

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CAREER OBJECTIVE

Space power electronics engineer with 3+ years of hands-on hardware experience, specialising in satellite EPS, PCDU and solar array design — including UMPPT/supercapacitor power conditioning around AZUR SPACE 3G30A triple-junction GaAs cells and complete 1U CubeSat development (OBC, EPS, Radio). Skilled in multi-layer PCB design, DC-DC converter and MPPT control, N-1 fault-tolerant power architectures, and full bench bring-up and validation, with current industry R&D in EV battery, BMS and motor-controller systems. Seeking to contribute to SOLLAB's space solar programme and build a long-term engineering career in Korea's space industry.

TECHNICAL SKILLS

- **Space Power Systems & EPS** — CubeSat-grade EPS architecture (1U/2U); PCDU design; UMPPT and MPPT control; triple-junction GaAs solar array design (AZUR SPACE 3G30A); hybrid Li-ion/supercapacitor storage; N-1 redundant fault-tolerant power distribution; OVP, OCP, reverse-polarity and thermal protection.
- **Power Electronics & Energy Conversion** — DC-DC converter design (buck, boost, buck-boost); switching regulator topology selection and control-loop compensation; Li-ion charging and protection; battery management systems including cell balancing, state-of-charge estimation and thermal management; on-board charger and motor-controller power stages.
- **PCB Design & Signal Integrity** — Multi-layer PCB design in EasyEDA Pro and KiCad; schematic capture; DRC/ERC closure; stack-up planning; EMI/EMC-aware layout; signal- and power-integrity routing; star grounding, ground-plane segmentation and chassis ground integration; Gerber/ODB++, BOM and pick-and-place generation; DFM hand-off to manufacturing.
- **Satellite Systems Engineering** — End-to-end 1U/2U CubeSat design; OBC, EPS, Radio, Antenna and Payload subsystem integration; CanSat development; low-power UHF communication architecture for LEO; LEO ground station design and automatic satellite tracking.
- **Embedded Systems & Firmware** — 8-bit and 32-bit MCU firmware (ATmega, ESP32); CAN, SPI, I2C and UART bus design and debugging; interrupt-driven architecture; low-power optimisation; sensor and peripheral integration; Python, C and C++.
- **Reliability & Test Engineering** — Reliability-oriented design and evaluation; N-1 redundancy architecture; failure-mode and root-cause analysis; functional validation; bench bring-up with oscilloscope, electronic load and programmable supply; SMD soldering and hot-air rework, harness assembly and crimping.
- **Modelling & Simulation** — LTspice and Proteus for converter and protection-circuit verification; MATLAB/Simulink for control-loop and system-level modelling; 3D printing, mechanical integration and enclosure prototyping.

WORK EXPERIENCE

Executive Electrical Engineer

Palki Motors Limited, Dhaka, Bangladesh | January 2026 – Present

- **Lead electrical R&D** on battery pack, BMS and motor-controller development for **Bangladesh's first indigenous electric vehicle** programme.
- Architect **BMS** functionality covering cell balancing, state-of-charge estimation, thermal management and layered fault protection, applying the same redundancy and protection principles used in satellite EPS design.
- Specify and validate DC-DC converter and on-board charger power stages; perform control-loop analysis, bench characterisation and root-cause investigation on **EV power hardware**.
- Work across mechanical, firmware and manufacturing teams to carry designs from schematic through prototype to production readiness.

Freelance PCB & Embedded Hardware Engineer

Fiverr & Freelancer.com (Self-Employed) | 2025 – Present

- Designed a spacecraft solar array and UMPPT/supercapacitor PCDU for **Sollab Co., Ltd. (South Korea)** using AZUR SPACE 3G30A triple-junction GaAs cells; ongoing remote R&D engagement.
- Designed a **40–50 V → 164 V / 45S LiFePO4 MPPT solar charge controller** — ESP32-driven boost converter with paralleled MOSFETs and isolated Hall-effect sensing, delivered with a 33-page datasheet-verified design report.
- Engineered a **4S→6S series-injection drone power module** (client under NDA) — eight-device inject FET bank with thermal analysis and a firmware-independent hardware SR-latch over-current trip.
- Designed a **36 × 36 mm 4-layer STM32G473 flight controller** — dedicated ground and power planes, star-ground via stitching, 90 Ω length-matched USB routing, released fab-ready.

PCB Design Engineer (R&D | Remote | Part-Time)

Robotry BD, Bangladesh | October 2023 – June 2025

- Designed multi-layer PCBs for commercial IoT products end-to-end — schematic capture, EMI/EMC-aware layout, DRC/ERC closure and full Gerber/BOM/pick-and-place release in EasyEDA Pro.
- Developed switching-regulator power rails and protection circuitry; integrated MCU firmware over I2C, SPI and UART with signal-level debugging.
- Delivered revision-controlled, production-ready designs with DFM documentation to manufacturing, supported by prototyping, SMD rework, bench testing and root-cause failure analysis.

RESEARCH & PUBLICATIONS

A Framework for Resilient Power Systems in CanSat-Class Reusable Small Satellites
1st Author | 26th International Middle East Power Systems Conference (MEPCON 2025), Aswan, Egypt |
DOI: ieeexplore.ieee.org/document/11360120

Reliability-Oriented Design and Evaluation of a Distributed CubeSat EPS with N-1 Redundancy and Hybrid Li-ion/Supercapacitor Storage
1st Author | 26th International Middle East Power Systems Conference (MEPCON 2025), Aswan, Egypt |
DOI: ieeexplore.ieee.org/document/11360261

A Low-Power UHF Communication Architecture for 1U CubeSats Operating in LEO
1st Author | 28th International Conference on Computer and Information Technology (ICCIT 2025), Cox's Bazar, Bangladesh
DOI: ieeexplore.ieee.org/document/11360261

Design and Development of an Indigenous Ground Station for LEO Orbit Satellites
2nd Author | IEEE International Conference on Next Generation Electronics (NEleX 2023), December 2023 |
DOI: ieeexplore.ieee.org/document/10421680

Electro-Mechanical Design Analysis and Implementation of an Indigenous Mars Rover (Rover-71)
3rd Author | IEEE International Conference on TENSYMP 2024, New Delhi, India |
DOI: ieeexplore.ieee.org/document/10752217

PROJECT EXPERIENCE

Undergraduate Thesis Project 2025: Design and Development of an Indigenous 1U Space-Grade CubeSat

- Architected and built a complete 1U space-grade CubeSat — full PCB layouts, fabrication and assembly of OBC, EPS, Radio, Antenna and Payload subsystems, followed by system validation and reliability testing; awarded **Best Thesis Award**.

CanSat Competition 2025 – USA | Electrical & Mechanical Lead

- Engineered system architecture and custom PCB design, fabricated and assembled a flight-ready CanSat, achieving 16th place among 40 international finalist teams at the AAS International CanSat Competition, USA (June 2025).

Mars Rover Competition 2025 – Turkey | Electrical Lead

- Developed electrical architecture and executed PCB integration and testing for Rover-71, securing 8th position globally (83.83%) among 27 international teams at ARC'25, Turkey.

1U Educational Grade CubeSat 2024 – University Campus | Electrical Lead

- Designed full system architecture, developed and fabricated custom PCBs, and built a fully functional 1U CubeSat with subsystem testing and validation (OBC, EPS, Radio).

Indigenous Automatic Tracking Ground Station 2023 – India | Electrical & Mechanical Lead

- Developed a custom low-cost mechanical architecture and integrated electronic control system, resulting in a fully functional automatic satellite tracking ground station.

EDUCATIONAL BACKGROUND

BSc in Aeronautical Engineering (Avionics)

Aviation & Aerospace University Bangladesh (AAUB) | 2022 – 2025 |

CGPA: 3.41 / 4.00

REFERENCES

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